# Band-1 Barrier Memo: P vs NP — Natural Proofs and Algebrization (Explicitly Not a Proof)

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## Band-1 Barrier Memo: P vs NP — Natural Proofs and Algebrization

**Honesty stamp (HARD).** This note does **not** claim that P $\neq$ NP. It does **not** claim that P = NP. It does **not** resolve the Clay Millennium Problem. It does **not** stamp Clay-closed, prize-ready, or Theory-of-Everything complete. It is a **barrier hygiene memo only**: a short, cited summary of two major relativization-style obstacles that constrain large classes of proof techniques for separating P from NP (and related complexity classes). Nothing here is a solution attempt.

### Abstract

Tonight's Band-1 collapse item **#10** asks only for a barrier memo. We summarize:

1. **Razborov–Rudich natural proofs** (1994/1997): under a widely believed cryptographic hypothesis, "natural" combinatorial lower-bound methods cannot separate P from NP (more precisely: cannot prove superpolynomial circuit lower bounds for NP).
2. **Aaronson–Wigderson algebrization** (2008/2009): even algebraic extensions of relativizing techniques (algebraic oracles / algebraic lifts) cannot settle P vs NP or several sibling questions; there exist algebrizing barriers in both directions for key statements.

**What this memo rules out:** not the truth of P $\neq$ NP, but large *method classes*. The open problem remains open. Preferred next work is non-natural, non-algebrizing technique search — or formalizing barrier *meta-statements* — not a Clay stamp.

### Problem reminder (status: OPEN)

The **P vs NP** question asks whether every language decidable in nondeterministic polynomial time is decidable in deterministic polynomial time. Equivalently: whether P = NP. Since Cook–Levin, Karp, and Levin, thousands of NP-complete problems are known; a polynomial algorithm for any one of them would collapse P = NP. No such algorithm is known, and no unconditional proof of inequality is known.

**Clay status:** Millennium Prize Problem; **unsolved**. This memo does not change that.

## Barrier 1 — Natural proofs (Razborov–Rudich)

### Statement (informal)

A combinatorial proof that a Boolean function $f_n : \{0,1\}^n \to \{0,1\}$ requires large circuits is called **natural** (roughly) if it proceeds by exhibiting a property $\Lambda$ of Boolean functions such that:

- **Constructivity:** membership in $\Lambda$ can be decided in quasipolynomial (or, in strong forms, polynomial) time given the truth table of a candidate function;
- **Largeness:** a non-negligible fraction of all Boolean functions satisfy $\Lambda$;
- **Usefulness:** every function in $\Lambda$ requires superpolynomial (or $2^{n^{\Omega(1)}}$) size circuits.

**Razborov–Rudich theorem (informal).** If there exist pseudorandom function families secure against quasipolynomial-size circuits (a standard cryptographic assumption, implied by e.g. the existence of exponentially hard one-way permutations under suitable parameters), then there is **no natural proof** that NP requires superpolynomial-size circuits.

### What natural proofs rule out

They rule out a broad class of **"combinatorial property + counting / density"** lower-bound strategies that:

- work for a large fraction of functions, and
- are themselves efficiently checkable from truth tables.

Many classical circuit lower-bound techniques for restricted models ($\mathrm{AC}^0$, $\mathrm{AC}^0[p]$, monotone circuits, etc.) *are* natural in this sense when lifted naively to the general Boolean circuit model. The barrier says: if cryptography is possible in the usual way, those styles of proof **cannot** finish the general P vs NP / NP-vs-P/poly separation.

They do **not** rule out:

- non-constructive proofs;
- proofs that exploit extremely sparse / non-large properties;
- proofs that are not efficiently decidable from truth tables;
- proofs in proof systems / frameworks outside the natural-proof template;
- lower bounds in highly restricted circuit classes where natural proofs *do* succeed (and historically have).

### Primary sources

- A. A. Razborov and S. Rudich, *Natural Proofs*, Proceedings of STOC 1994; journal version *Journal of Computer and System Sciences* **55** (1997), 24–35.
- Survey context: S. Arora and B. Barak, *Computational Complexity: A Modern Approach*, Cambridge, 2009, Ch. 23; L. Trevisan survey notes on natural proofs.

## Barrier 2 — Algebrization (Aaronson–Wigderson)

### Motivation

**Baker–Gill–Solovay relativization** (1975) showed that there exist oracles $A, B$ with $\mathrm{P}^A = \mathrm{NP}^A$ and $\mathrm{P}^B \neq \mathrm{NP}^B$. Any proof technique that "relativizes" (remains valid after adding an arbitrary oracle) therefore cannot settle P vs NP.

Algebraic techniques (arithmetization: low-degree polynomial extensions of Boolean functions; IP=PSPACE; MIP; PCP) appeared to escape pure relativization. Aaronson–Wigderson asked whether a refined barrier still applies once oracles are allowed to answer **algebraic** queries (low-degree extensions / polynomial-oracle access).

**Statement (informal)**

A complexity-class inclusion/separation **algebrizes** if it continues to hold in the presence of algebraic oracles (oracles that provide values of low-degree extensions of the oracle language, not merely Boolean membership).

**Aaronson–Wigderson (selected consequences, informal):**

- Any proof that P = NP that algebrizes is impossible in the same sense that relativizing proofs are blocked: there are algebraic oracles relative to which P $\neq$ NP (and vice versa for the other direction on related statements).
- More sharply: for several central questions (P vs NP, NP vs coNP, permanent vs #P completeness subtleties, etc.), both an algebrizing proof of equality *and* an algebrizing proof of separation are ruled out by suitable algebraic-oracle worlds.
- Famous non-relativizing results such as IP = PSPACE *do* algebrize, which explains why arithmetization escaped Baker–Gill–Solovay yet still sits inside a larger barrier envelope for P vs NP.

**What algebrization rules out**

It rules out proof techniques that remain valid under algebraic oracle extensions — including many arithmetization-friendly arguments that escape classical relativization but still "look local" with respect to polynomial extensions.

It does **not** rule out:

- non-algebrizing techniques (e.g. some interactive / cryptographic / geometric complexity theory / proof-complexity routes, depending on formalization);
- unconditional circuit lower bounds obtained by methods outside both natural and algebrizing templates;
- the actual Boolean (oracle-free) truth of P $\neq$ NP or P = NP.

**Primary sources**

- S. Aaronson and A. Wigderson, *Algebrization: A New Barrier in Complexity Theory*, Proceedings of STOC 2008; journal version *ACM Transactions on Computation Theory* **1**(1) (2009), Article 2.
- Background: T. Baker, J. Gill, and R. Solovay, *Relativizations of the* P = ?NP *question*, *SIAM Journal on Computing* **4**(4) (1975), 431–442.
- Context survey: S. Aaronson, $P \stackrel{?}{=} NP$, in *Open Problems in Mathematics* (Nash–Rassias, eds.), Springer, 2016.

## Comparative ledger (what is blocked)

| Barrier | Year | Rules out (method class) | Does **not** settle |
|---|---|---|---|
| Relativization (Baker–Gill–Solovay) | 1975 | Techniques valid under arbitrary Boolean oracles | Truth of P vs NP |
| Natural proofs (Razborov–Rudich) | 1994/97 | Large + constructive combinatorial circuit lower bounds (under crypto assumptions) | Non-natural / non-large proofs |
| Algebrization (Aaronson–Wigderson) | 2008/09 | Techniques that survive algebraic-oracle lifts | Non-algebrizing techniques |

**Takeaway for Band-1:** a serious attack on P vs NP must explain why it is **non-relativizing**, **non-natural**, and **non-algebrizing** — or must work in a restricted model where these barriers do not apply. Claiming a Clay-grade resolution without that hygiene is out of house rules.

### Optional Lean-formalizable meta-statement (suggestion only)

Not implemented tonight. A useful *hygiene* formalization target (meta, not a Clay proof):

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/-- Barrier hygiene, not a P vs NP resolution.
    Informal target for a future Lean sketch:

    axiom RelativizingTechnique : Technique → Prop
    axiom NaturalTechnique : Technique → Prop
    axiom AlgebrizingTechnique : Technique → Prop

    /-- No technique that is simultaneously "only relativizing"
        can decide PVsNP. (Baker-Gill-Solovay as a meta-axiom.) -/
    axiom no_relativizing_settles_PvsNP :
      forall t, RelativizingTechnique t → ¬ SettlesPvsNP t

    /-- Under a crypto hypothesis CryptoHyp, no natural technique
        yields NP notsubseteq P/poly. (Razborov-Rudich as a meta-axiom.) -/
    axiom no_natural_NP_circuit_lower_bound :
      CryptoHyp → forall t, NaturalTechnique t → ¬ ProvesNPnotInPpoly t

    /-- Algebrizing techniques do not settle P vs NP.
        (Aaronson-Wigderson as a meta-axiom.) -/
    axiom no_algebrizing_settles_PvsNP :
      forall t, AlgebrizingTechnique t → ¬ SettlesPvsNP t

    /-- House rule: we do NOT assert SettlesPvsNP for any concrete t. -/
-/
```

This is a **suggested interface** for a future `Formal-Conjectures`-style barrier module. It encodes the barriers as axioms about *techniques*, not as a proof that P $\neq$ NP. Implementing even a

toy `Technique` inductive with tagged constructors (`relativizing`, `natural`, `algebrizing`, `other`) would already be a valuable Lean hygiene artifact.

### What we are explicitly not doing

- Not exhibiting an algorithm placing an NP-complete problem in P.
- Not proving a superpolynomial circuit lower bound for an NP language in the general model.
- Not claiming geometric complexity theory (GCT), proof complexity, or fine-grained complexity has closed the problem.
- Not requesting or implying a Clay prize submission.
- Not updating any "solved" bit in mission-control open-problem ledgers.

### Band-1 outcome

| Field | Value |
|---|---|
| Item | Tonight's ten #10 |
| Slug | `pvsnp_barrier_memo` |
| Claim type | Barrier summary / method hygiene |
| P vs NP status | **OPEN** (unchanged) |
| Clay stamp | **None** |
| Primary barriers cited | Razborov–Rudich natural proofs; Aaronson–Wigderson algebrization (plus BGS relativization background) |
| Lean artifact tonight | Optional meta-statement sketched only — not checked-in as a `.lean` file |

### Paths

- Markdown: `/workspace/mission-control/papers/2026-09-11_Band1_PvsNP_barrier_memo.md`
- PDF (shelf): `/workspace/DOWNLOAD_PDFs/2026-09-11_Band1_PvsNP_barrier_memo.pdf`
- PDF (papers twin): `/workspace/mission-control/papers/2026-09-11_Band1_PvsNP_barrier_memo.pdf`

### References (compact)


1. T. Baker, J. Gill, R. Solovay. Relativizations of the P=?NP question. *SIAM J. Comput.* 4(4):431–442, 1975.
2. A. A. Razborov, S. Rudich. Natural Proofs. *J. Comput. System Sci.* 55:24–35, 1997 (STOC 1994).
3. S. Aaronson, A. Wigderson. Algebrization: A New Barrier in Complexity Theory. *ACM Trans. Comput. Theory* 1(1):2, 2009 (STOC 2008).
4. S. Arora, B. Barak. *Computational Complexity: A Modern Approach*. Cambridge University Press, 2009.
5. S. Aaronson. P $\stackrel{?}{=}$ NP. In *Open Problems in Mathematics*, Springer, 2016.


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*ZoraASI / C.M. Baird — Band-1 full-dash, 2026-09-11. Barrier memo only. Not a proof.*